Test Case / Original Request

Carry out PTC-009 as a real k6 test. Monitor the run, assess security for its configured API, analyze the current job, and create a PDF that links to the native report and raw results.

REAL Execution Outcome

Execution status: COMPLETED
Runner status: completed
Outcome details: REAL k6 execution completed. Threshold breaches, if any, are retained as performance findings.
Thresholds passed: True

Executive Summary

Application: Unknown
Scenario: Customer Directory API — Load Check
Job ID: JOB-5b3101bb77334ec6b007e6c3deffcd9b
Overall Result: PASS

All tests completed successfully. API performance was stable under load, with low response times, no errors, and no threshold breaches. System resources were within normal limits, with moderate memory and low CPU usage observed. Security controls for the API passed live observation, though the result is limited to header presence in a single HEAD request. No issues or anomalies were detected across performance, infrastructure, or security aspects.

Execution Profile and Artifacts

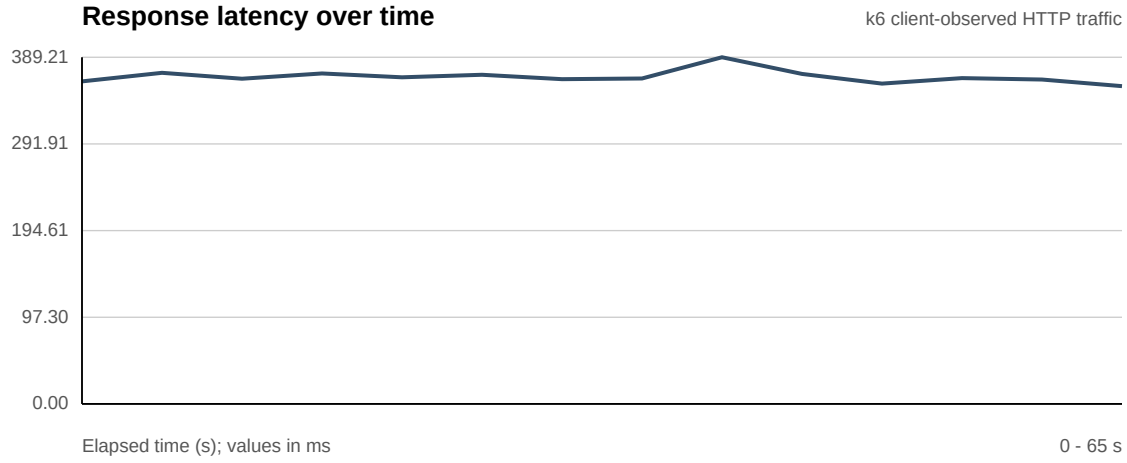
Execution mode: REAL
Workload profile:
- tool: k6
- target_url: https://dummyjson.com
- endpoint: /users/1
- method: GET
- users: 120
- max_rps: 12
- duration_seconds: 60
- ramp_up_seconds: 5
- expected_status_code: 200
- p95_threshold_ms: 5000
Artifact directory: src/generated_scripts/k6 native reports/JOB-5b3101bb77334ec6b007e6c3deffcd9b
Artifact references:
- aligned_time_series: src/generated_scripts/k6 native reports/JOB-5b3101bb77334ec6b007e6c3deffcd9b/k6_PTC_009_Customer_Directory_API_Load_Check_aligned_time_series.csv
- generated_script: src/generated_scripts/k6 native reports/JOB-5b3101bb77334ec6b3deffcd9b/k6_PTC_009_Customer_Directory_API_Load_Check.js
- k6_summary: src/generated_scripts/k6 native reports/JOB-5b3101bb77334ec6b007e6c3deffcd9b/k6_PTC_009_Customer_Directory_API_Load_Check_summary.json
- measurement_metadata: src/generated_scripts/k6 native reports/JOB-5b3101bb77334ec6b007e6c3deffcd9b/k6_PTC_009_Customer_Directory_API_Load_Check_measurement_metadata.json
- monitoring_samples: src/generated_scripts/k6 native reports/JOB-5b3101bb77334ec6b007e6c3deffcd9b/k6_PTC_009_Customer_Directory_API_Load_Check_runtime_samples.csv

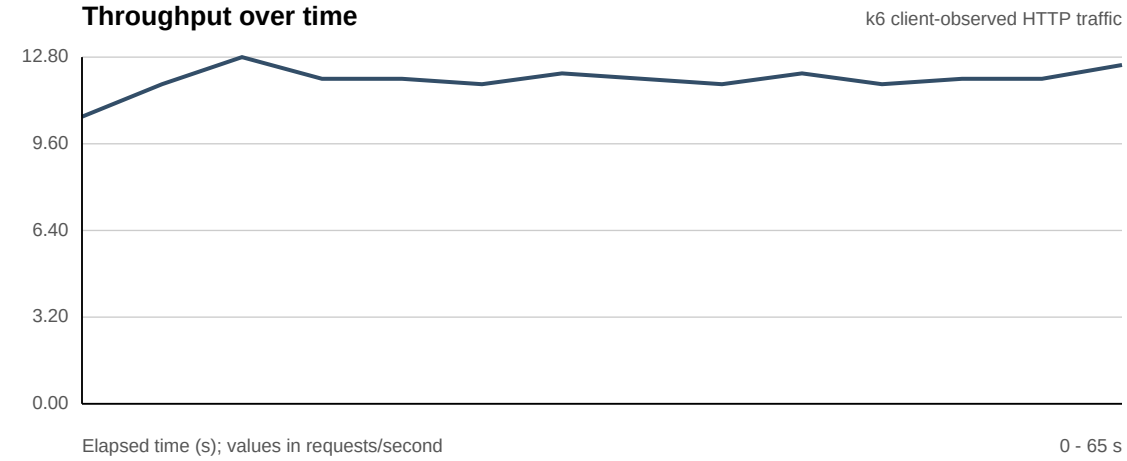
- Native k6 HTML report
- Native report + raw results ZIP
- normalized_metrics: src/generated_scripts/k6 native reports/JOB-5b3101bb77334ec6b007e6c3deffcd9b/k6_PTC_009_Customer_Directory_API_Load_Check_normalized_metrics.json
- Raw k6 results (JSON)
- runner_log: src/generated_scripts/k6 native reports/JOB-5b3101bb77334ec6b007e6c3deffcd9b/k6_PTC_009_Customer_Directory_API_Load_Check_runner.log
- time_series: src/generated_scripts/k6 native reports/JOB-5b3101bb77334ec6b007e6c3deffcd9b/k6_PTC_009_Customer_Directory_API_Load_Check_time_series.csv

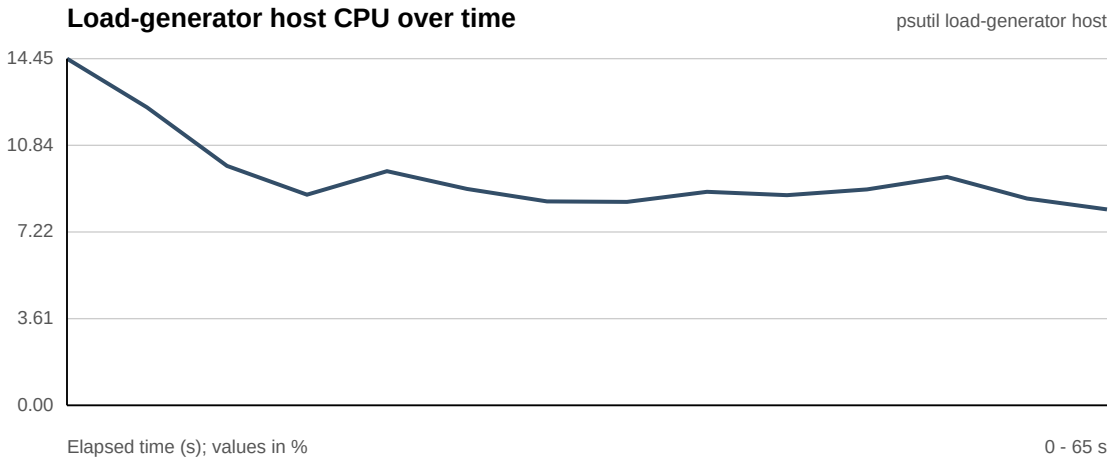
The interactive native HTML dashboard is linked as a separate artifact and is not embedded in this PDF.

Time-Series Charts (REAL Measurements)

Charts show measured values in five-second aligned buckets. Host CPU is from the load-generator machine, not the target.







Agent Activity Summary

Agent	*Summary*
Test Execution	Load test executed on /users/1 (Unknown - Customer Directory API — Load Check). Results: 812 total requests, 812 successful, 0 failed. Avg response time: 367.6255874384237 ms; 95th percentile: 450.38495 ms; Max: 781.174 ms. Throughput: 11.959729880957974 TPS; Error rate: 0.0%.
Monitoring	CPU usage: 9.57% average, 18.2% peak; memory: 67.99% average, 70.0% peak; DB connections: not collected, TPS: 11.959729880957974, Response time: 367.6255874384237 ms. Data source and scope: REAL: psutil load-generator host samples and k6 client-observed metrics. Measurement provenance: cpu_usage=psutil (load_generator_host); cpu_peak_usage=psutil (load_generator_host); memory_usage=psutil (load_generator_host); memory_peak_usage=psutil (load_generator_host); response_time=k6 (load_generator_observed_http_traffic); tps=k6 (load_generator_observed_http_traffic); db_connections=not_collected (No target-specific database telemetry provider is configured.). No alerts triggered.
Security Validation	Status: PASS. Security score: 100%. Live security check: completed. Issue: Scope limitation: configuration validator outcomes describe profile declarations; any live result is a separate single-request observation and does not prove control effectiveness. Issue: Live HEAD observation completed with HTTP status 200; response headers are observations only, not proof of control effectiveness. Control coverage: {'requested_count': 1, 'assessed_count': 1, 'not_assessed_count': 0} Control assessment evidence: - external_surface: assessed; basis=live_observation; evidence=['live-observation:JOB-5b3101bb77334ec6b007e6c3deffcd9b'] Evidence references: live-observation:JOB-5b3101bb77334ec6b007e6c3deffcd9b

Agent	*Summary*
Analysis	All tests completed successfully. API performance was stable under load, with low response times, no errors, and no threshold breaches. System resources were within normal limits, with moderate memory and low CPU usage observed. Security controls for the API passed live observation, though the result is limited to header presence in a single HEAD request. No issues or anomalies were detected across performance, infrastructure, or security aspects.
Reporting	A PDF report was generated, including all findings, recommendations, and an executive summary.

Per-API Findings (Test Execution)

- /users/1 [PASS / low]
avg=367.6255874384237ms p95=450.38495ms max=781.174ms throughput=11.959729880957974 req/s error_rate=0.0% http_status=200

Infrastructure Health (Monitoring)

Status: PASS (monitoring_agent reported: OK)
CPU usage: 9.57% Memory usage: 67.99% DB connections: None TPS: 11.959729880957974
Response time (monitoring): 367.6255874384237ms

Security Findings (Security Agent)

Status: PASS Security score: 100%
Security result job ID: JOB-5b3101bb77334ec6b007e6c3deffcd9b.
Live security check: completed.
Issues:
- Scope limitation: configuration validator outcomes describe profile declarations; any live result is a separate single-request observation and does not prove control effectiveness.
- Live HEAD observation completed with HTTP status 200; response headers are observations only, not proof of control effectiveness.
Control coverage: {'requested_count': 1, 'assessed_count': 1, 'not_assessed_count': 0}
Requested controls: external_surface
Assessed controls: external_surface
Control evidence and provenance:
- external_surface: assessed; basis=live_observation; evidence fields=['target_origin', 'endpoint', 'method', 'response_status_code', 'observed_security_headers']; references=['live-observation:JOB-5b3101bb77334ec6b007e6c3deffcd9b']
Evidence references: live-observation:JOB-5b3101bb77334ec6b007e6c3deffcd9b
Live observation: HEAD /users/1; HTTP 200; elapsed 567.2621669946238 ms; requests sent 1.
Observed response headers (observations only):
- strict-transport-security: max-age=15552000; includeSubDomains
- x-content-type-options: nosniff
- x-frame-options: SAMEORIGIN

Recommended Next Steps

All dimensions passed -- no action required.